

A disproof of Conjecture 00000007794

Reeve tetrahedra break log-concavity of Ehrhart coefficients

1 The conjecture

Conjecture 00000007794 reads (English version):

Definition: Log-concavity of Ehrhart coefficients: $L(P, k)$ are the coefficients of the Ehrhart series of a lattice polytope P . Conjecture: The log-concavification satisfies $L(P, k)^2 \geq L(P, k-1)L(P, k+1)(1 + c/k^2)$ for universal c , i.e. log-concavity with explicit stability margin; equality holds at the simplex; and the real-rootedness of the h^* -polynomial has an independent characterization with all roots of $h^*(P)$ in the annulus $\{z : r \leq |z| \leq 1\}$ for explicit lower bound r .

The Chinese version writes the Ehrhart series explicitly as $\sum L(P, k)t^k$ (“ge duobaoxing P de Ehrhart jishu $\sum L(P, k)t^k$ de xishu $L(P, k)$ ”: “the coefficients $L(P, k)$ of the Ehrhart series $\sum L(P, k)t^k$ of the lattice polytope P ”), and the margin as “c pushi” (“ c universal”).

Result. The first clause is false for every constant $c > -1$, which covers every positive “stability margin” and plain log-concavity ($c = 0$). The Reeve tetrahedra violate it at $k = 1$. Since the conjecture is a conjunction, it is false. Everything is formalized in Lean 4 with Mathlib (Section 4).

2 Definitions, reading and conventions

Definition 1. A *lattice polytope* is the convex hull of a nonempty finite set of lattice points. For $k \in \mathbb{N} = \{0, 1, 2, \dots\}$, $L(P, k) = \#(kP \cap \mathbb{Z}^d)$; these are the coefficients of the Ehrhart series $\sum_{k \geq 0} L(P, k)t^k$. Source: Wikipedia, *Ehrhart polynomial* (https://en.wikipedia.org/wiki/Ehrhart_polynomial): $L(P, t)$ is “the number of lattice points contained in the polytope tP ”, and the Ehrhart series is $\text{Ehr}_P(z) = \sum_{t \geq 0} L(P, t)z^t$.

Definition 2 (Reeve tetrahedron). For an integer $r \geq 1$, $T_r = \text{conv}\{(0, 0, 0), (1, 0, 0), (0, 1, 0), (1, 1, r)\} \subset \mathbb{R}^3$. Source: Wikipedia, *Reeve tetrahedra* (https://en.wikipedia.org/wiki/Reeve_tetrahedra): the vertices are as above, “where r is a positive integer”.

Reading. The first clause says: there is a constant c , independent of P and k , such that for every lattice polytope P and every $k \geq 1$,

$$L(P, k)^2 \geq L(P, k-1)L(P, k+1)(1 + c/k^2).$$

The range $k \geq 1$ is the one where c/k^2 and $L(P, k-1)$ are both defined. We refute this for every real $c > -1$, in dimension 3. A statement for all dimensions in particular covers $d = 3$.

Conventions. (i) $L(P, 0) = \#(0 \cdot P \cap \mathbb{Z}^3) = \#\{0\} = 1$; this is computed from the definition (Lean: `ehrhart_zero`), and it agrees with the constant term 1 of the Ehrhart series. (ii) kP is the

pointwise dilation $\{kx : x \in P\}$. (iii) Counting uses Mathlib's `Set.ncard`, which is 0 on infinite sets. Our lower bound for $L(T_r, 2)$ therefore includes a proof that the lattice points of $2T_r$ form a finite set. (iv) $k - 1$ is natural subtraction, which is exact for $k \geq 1$. (v) Full-dimensionality is not part of the definition of a lattice polytope; anyway T_r is a genuine tetrahedron of volume $r/6$.

3 Proof

Lemma 3 (facet inequalities). *For $s \geq 0$, every $x \in sT_r$ satisfies $0 \leq x_3$, $x_3 \leq rx_1$, $x_3 \leq rx_2$ and $r(x_1 + x_2) \leq sr + x_3$.*

Proof. For $s = 1$, the four vertices satisfy these inequalities. The set they define is an intersection of half-spaces, hence convex, so it contains the convex hull T_r . Multiply by $s \geq 0$. $\square$

Lemma 4. *$L(T_r, 1) \leq 4$ for $r \geq 1$: every lattice point of T_r is a vertex.*

Proof. Let $(x, y, z) \in \mathbb{Z}^3 \cap T_r$. If $z = 0$: $rx \geq 0$ and $ry \geq 0$ give $x, y \geq 0$, and $r(x + y) \leq r$ gives $x + y \leq 1$; so $(x, y) \in \{(0, 0), (1, 0), (0, 1)\}$. If $z > 0$: $rx \geq z > 0$ gives $x \geq 1$, and likewise $y \geq 1$. Then $2r \leq r(x + y) \leq r + z$ gives $z \geq r$, and $2z \leq r(x + y) \leq r + z$ gives $z \leq r$. So $z = r$, and $r(x + y) \leq 2r$ forces $x = y = 1$. $\square$

Lemma 5. *$L(T_r, 2) \geq r + 1$ for $r \geq 1$.*

Proof. Finiteness. By Lemma 3 with $s = 2$, a lattice point (x, y, z) of $2T_r$ has $x, y \geq 0$, $2z \leq r(x + y) \leq 2r + z$ (so $z \leq 2r$), and $r(x + y) \leq 4r$ (so $x + y \leq 4$). It therefore lies in the box $[0, 4] \times [0, 4] \times [0, 2r]$. *Points.* For $0 \leq z \leq r$, put $w = z/(2r) \in [0, \frac{1}{2}]$. The convex combination with weights $w, \frac{1}{2} - w, \frac{1}{2} - w, w$ of the vertices $(0, 0, 0), (1, 0, 0), (0, 1, 0), (1, 1, r)$ is $(\frac{1}{2}, \frac{1}{2}, \frac{z}{2}) \in T_r$. So $(1, 1, z) \in 2T_r$. These are $r + 1$ distinct lattice points. $\square$

Theorem 6. *Let $c > -1$, and let $r \geq 1$ satisfy $(r + 1)(1 + c) > 16$, for example $r = \lceil 16/(1 + c) \rceil + 1$. Then*

$$L(T_r, 1)^2 < L(T_r, 0) L(T_r, 2) (1 + c/1^2).$$

Hence no $c > -1$ satisfies the first clause. In particular T_{16} violates plain log-concavity: $L(T_{16}, 1)^2 \leq 16 < 17 \leq L(T_{16}, 0)L(T_{16}, 2)$.

Proof. $L(T_r, 1)^2 \leq 16 < (r + 1)(1 + c) \leq 1 \cdot L(T_r, 2) \cdot (1 + c)$, by Lemmas 4 and 5, $L(T_r, 0) = 1$ and $1 + c > 0$. $\square$

Cross-check, not used in the proof. Wikipedia (*Reeve tetrahedra*) states: “The Ehrhart polynomial of the Reeve tetrahedron Tr of height r is” $\frac{r}{6}t^3 + t^2 + (2 - \frac{r}{6})t + 1$. This gives $L = 1, 4, r + 9, 4r + 16, \dots$. A brute-force count confirmed these values for $r \in \{1, 2, 8, 9, 13\}$ and $k \leq 3$. With the exact value $r + 9$, plain log-concavity already fails at $r = 8$ ($16 < 17$).

4 Lean formalization

File `lean/Conjecture7794/Basic.lean` (251 lines, namespace `C7794`; only import `Mathlib`). Definitions: `E3 = Fin 3 -> R`, `toR`, `IsLatticePolytope` (convexHull of the image of a nonempty `Finset` of integer points), `latticePts`, `ehrhart P k = (latticePts ((k : R) . P)).ncard`, `vZ`, `V`, `reeve`, `H`. Lemmas: `reeve_isLatticePolytope`, `convex_H`, `reeve_subset_H`, `smul_reeve_subset` (Lemma 3); `ehrhart_zero`; `latticePts_one`, `ehrhart_one_le` (Lemma 4); `latticePts_two_finite`, `mem_two_reeve`, `ehrhart_two_ge` (Lemma 5); `reeve_violates` (Theorem 6). Main theorem (ASCII rendering):

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theorem conjecture_7794_false :
  not (exists c : R, -1 < c and forall P : Set E3, IsLatticePolytope P ->
    forall k : N, 1 <= k ->
      (ehrhart P (k - 1) : R) * ehrhart P (k + 1) * (1 + c / (k : R) ^ 2)
        <= (ehrhart P k : R) ^ 2) and
  (forall c : R, -1 < c -> exists r : N, 1 <= r and
    IsLatticePolytope (reeve r) and
    (ehrhart (reeve r) 1 : R) ^ 2 <
      (ehrhart (reeve r) 0 : R) * ehrhart (reeve r) 2
        * (1 + c / ((1 : N) : R) ^ 2)) and
  (ehrhart (reeve 16) 1) ^ 2
    < ehrhart (reeve 16) 0 * ehrhart (reeve 16) 2

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Axiom audit. #print axioms C7794.conjecture_7794_false reports [propext, Classical.choice, Quot.sound]. There is no sorry and no native_decide. The file compiles under lake build and with -DwarningAsError=true.

5 Scope

- Refuted: the first clause for every universal $c > -1$ (all positive margins and $c = 0$), with k ranging over $k \geq 1$, in dimension 3. The planar quadrilaterals suggested by an automated filter are *not* used: by Pick’s theorem, lattice polygons satisfy plain log-concavity at $k = 1$.
- Not refuted: $c \leq -1$. Then $1 + c/k^2 \leq 0$ at $k = 1$, so the inequality is not a stability margin at all. Also not refuted: a reading restricted to $k \geq 2$. Reeve tetrahedra satisfy plain log-concavity at $k = 2$ ($(r + 9)^2 > 4(4r + 16)$), so they say nothing about small $c > 0$ there.
- The clauses “equality holds at the simplex” and the h^* annulus are not addressed. Note that T_r is itself a lattice simplex.